

Comparative Study of Parameter-Efficient Fine-Tuning on Low-Resource Vision-Language Modelling: Burmese-CLIP

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Abstract

Contrastive Language-Image Pre-Training (CLIP) is a powerful vision-language architecture, but fine-tuning it requires substantial computational and data resources. We present the Burmese-CLIP, trained on 20,000 machine-translated image-caption pairs using Low Rank Adaptation (LoRA). Our Burmese-CLIP achieves approximately a seven-fold improvement in Mean Reciprocal Rank (MRR) for text-to-image retrieval over zero-shot NLLB-CLIP. We further conduct a systematic comparison of three PEFT methods, LoRA, DoRA, and AdaLoRA, under identical training conditions to understand their behaviours in a low-resource cross-modal fine-tuning environment. LoRA at rank 32, the model achieves the strongest retrieval quality (MRR@1 of 0.2214), while DoRA at rank 16 retains approximately 86% of this performance with nearly half the trainable parameters, suggesting a favourable efficiency trade-off. AdaLoRA under-performs across all metrics, which we hypothesise that it might be due to the aggressive pruning of attention projection layers, which are critical for cross-modal semantic alignment.

Keywords: Low-Resource Language, Burmese-CLIP, Parameter-Efficient Fine-Tuning(PEFT), Contrastive Learning

1 Introduction

Contrastive Language-Image Pre-Training (CLIP) is a powerful architecture that uses text and visual transformers to encode information into a shared representation space using contrastive learning (Radford et al., 2021; Dosovitskiy et al., 2020; Vaswani et al., 2017). Trained on roughly 400 million image-text pairs from the web (Radford et al., 2021), CLIP achieves outstanding results on various downstream tasks such as zero-shot classification (Thengane et al., 2022; Li et al., 2022), text-image and image-text extraction and more complex tasks such as semantic segmentation (Lin et al., 2023).

CLIP has been adapted to other languages and domains. The Italian CLIP (Bianchi et al., 2021) was adapted by using 1.4 million Italian image-text pairs, while the Chinese CLIP (Yang et al., 2023) scales the approach to 200 million native pairs. Multilingual CLIP (mCLIP) (Zhang et al., 2023b) pursues a broader strategy, mapping dozens of languages into a single shared visual space, and AltCLIP (Chen et al., 2022) uses a teacher-student framework over 2 million image-caption pairs to support cross-lingual retrieval. For low-resource languages, NLLB-CLIP (Visheratin, 2023b) has been developed by replacing the CLIP text encoder with ‘No Language Left Behind’ (NLLB) text encoder achieving excellent results in low-resource languages. Domain-specific variants follow a similar pattern - FashionCLIP (Cartella et al., 2023) fine-tuned CLIP for the retail sector using around 700,000 curated image-text pairs.

In this paper, we explore how different Parameter Efficient Fine Tuning (PEFT) methods perform in fine-tuning CLIP in the low-resource language environment using a small amount of machine-translated data. Therefore, for this experiment, we chose Burmese as our target language. It is due to its unique computational and linguistic challenges such as lack of robust support in ASCII-based systems online for Burmese scripts (Htet and

Dras, 2025). We introduce Burmese-CLIP which is trained on 20,000 machine-translated image-caption pairs from LAION-COCO-NLLB dataset using various Low-Rank Adaptation methodologies (Visheratin, 2023a). We replaced the text encoder with multilingual XLM-Roberta-Base (XLM-R) (Conneau et al., 2020), which was chosen after a token fertility analysis. Our results indicate that Burmese-CLIP performs better than zero-shot NLLB-CLIP in terms of Mean Reciprocal Rank (MRR) for text-to-image retrieval task, while updating only 4.3% of trainable parameters.

We also investigate the performance of different LoRA methods in developing Burmese-CLIP. We utilised Decomposed Low-Rank Adaptation (DoRA) (Liu et al., 2024) with two different ranks and Adaptive Low-Rank Adaptation (AdaLoRA) (Zhang et al., 2023a), alongside LoRA with 32 rank. The evaluation result reveals LoRA provides the highest with MRR@1 of 0.2214, while DoRA ($r=16$) retains 86% of the retrieval accuracy while training nearly half of the trainable parameters of LoRA. Meanwhile, AdaLoRA with dynamic target rank of 16 under-performs across the board. Our analysis reveals the importance-scoring mechanism prematurely eliminated key attention layers, confirming the AdaLoRA’s bias toward FFN layers, which works against CLIP due to its reliance upon query and key projections (Zhang et al., 2023a; Chung et al., 2025; Wang et al., 2024). To our knowledge, this is the first systematic PEFT study for Burmese-specific CLIP adaptation.

2 Related Work

2.1 Low-Rank Adaptation (LoRA)

Fine-tuning Large Language Models is computationally expensive and data-intensive (Han et al., 2024; Szep et al., 2025; Hu et al., 2022). Low-Rank Adaptation (LoRA) addresses this by freezing pre-trained weights and injecting trainable low-rank matrices as weight updates (Hu et al., 2022; Han et al., 2024; Ding et al., 2023a,b):

$$\Delta W = BA, \quad W_{\text{new}} = W_0 + BA, \quad (1)$$

with $A \in \mathbb{R}^{d \times r}$, $B \in \mathbb{R}^{r \times d}$, and $r \ll d$, reducing trainable parameters to 0.1–3% of the original model while preserving performance relative to full fine-tuning (Han et al., 2024; Hu et al., 2022).

Several LoRA variants have refined this further (Han et al., 2024). DoRA decomposes pre-trained weights into magnitude and direction components, optimising each separately (Liu et al., 2024). AdaLoRA treats low-rank matrices as an SVD and dynamically allocates the parameter budget via an importance-scoring metric (Zhang et al., 2023a), concentrating learning on the layers most critical to the downstream task.

2.2 Low-Resource Language - Burmese Language

Although Myanmar (Burma) has a population of roughly 54-55 million (World Bank, 2023; United Nations Department of Economic and Social Affairs, 2024), Burmese has a limited machine-readable digital footprint (Htet and Dras, 2025), shaped by political, socio-economic, and infrastructural constraints that leave text resources sparse, fragmented, and domain-narrow (Htet and Dras, 2025). This is compounded by encoding fragmentation. The Burmese script lacked robust ASCII support (Htet and Dras, 2025), and although Unicode is now the standard, much online Burmese texts remain in the legacy Zawgyi font system, introducing substantial corpus noise (Oo et al., 2020; Htet and Dras, 2025; Liao, 2017).

Burmese orthography poses further challenges. It does not mark word boundaries with whitespace, and meaning derives from syllable combinations, so naive segmentation can alter semantic meaning (Thet et al., 2008; Maung and Mikami, 2008; Ding et al., 2019). Thus, the chosen tokeniser must minimise out-of-vocabulary fragmentation to avoid such semantic hallucinations (Ding et al., 2019; Thet et al., 2008; Htet and Dras, 2025).

Recent Burmese NLP works include MyanmarGPT (Si Thu, 2024), trained on a curated multi-domain corpus with syllable-aware tokenisation, and BURMESE-SAN (Aung et al., 2026), a benchmark for evaluating LLMs on Burmese-specific tasks. SEA-LION (Ng et al., 2025) similarly argues for targeted regional pre-training on underrepresented Southeast Asian languages. Our work extends this landscape from text-only modelling toward multi-modal.

3 Pre-trained Multilingual Model Selection

Due to unique challenges of the Burmese language, we conducted a fertility analysis to select the most suitable pre-trained multilingual model for Burmese-CLIP. We defined fertility as the ratio of produced tokens to input words for a given text sequence (Rust et al., 2021; Nayeem et al., 2025). Lower fertility indicates that the tokeniser encodes more information per token, where high fertility corresponds to fragmentation into byte-level or encoding-like artefacts that carry minimal semantic content (Nayeem et al., 2025). A total of 47 Burmese sentences were used for this analysis. These sentences can be categorised into machine-translated sentences, similar to the training data, simple sentences such as a common greeting, weather description, and more complex sentences curated by the authors from Burmese news websites such as BBC Burmese, VOA Burmese with topics ranging from politics, economics, education and sport. Results are summarised in Table 1, and XLM-Roberta-Base and mT5-Base outperform the rest. They both produce a clean and coherent tokenisation of Burmese language due to inclusion of Burmese languages as well as employment of SentencePiece tokeniser (Conneau et al., 2020; Xue et al., 2021). XLM-Roberta-Base was selected as the text encoder due to its architecture which aligns with the CLIP model (Radford et al., 2021) and wide applications in many other clips models (Zhang et al., 2023b; Carlsson et al., 2022).

Model Name	Average Fertility Score
Mistral 7B v0.1	1.546
BLOOMZ-560M	1.682
ByT5-small	2.875
mGPT	0.730
mBERT	0.616
mT5-base	0.328
XLM-Roberta-Base	0.325

Table 1: Fertility Analysis of Different Multilingual Models

4 Methodology

4.1 Training & Evaluation Dataset

The required dataset was sourced from `laion-coco-nllb` dataset (Visheratin, 2023a). English captions have been machine-translated into over 200 languages using the NLLB-200 model (Visheratin, 2023a). We filtered the dataset for entries carrying Burmese captions with the aesthetic score threshold ($\text{score} \geq 5.5$) to retain visually informative images to select images with stronger compositional and visual quality. The final dataset comprises approximately 20,000 image-caption pairs. 2,000 image-caption pairs were held out for evaluation. Manual inspection of a random sample of 20 pairs found 18 correctly aligned Burmese captions. The dataset uses Burmese Unicode and therefore, it was not needed to use Unicode converter. No other normalisation practices were performed on the dataset. It is noted that due to a machine translation, some semantic drift and translation artefacts are noted.

4.2 Training

Burmese-CLIP was trained using `VisionTextDualEncoderModel`, which supports independent visual and textual encoders that produce embeddings into a shared projection space. The visual encoder is OpenAI’s CLIP ViT-B/32 that produces a 512-dimensional image embedding, and is frozen throughout all training phases. This is because the visual representation transfers well across different languages (Bianchi et al., 2021; Visheratin, 2023b). The text encoder is XLM-Roberta-Base with 12-layer bidirectional transformer producing a 768-dimensional [CLS] representation projected into the shared 512-dimensional embedding space via a learned linear projection head.

The model is trained using the symmetric contrastive loss from the original CLIP formulation. For a batch of N image-caption pairs, the losses are cross-entropy losses over the rows and columns of the $N \times N$ similarity matrix respectively, with ground-truth labels on the diagonal (Radford et al., 2021; Vishnath, 2023b; Bianchi et al., 2021).

Training follows a two-phase training motivated by prior work of Italian-CLIP adaptation. In phase 1 (3 epochs), both encoders are frozen and only the projection layers are updated. In Phase 2 (7 epochs), LoRA adaptors are applied to only the text encoder. The rationale is that the premature fine-tuning of the text encoder risks destabilising the shared embedding space before a coherent alignment has been established (Bianchi et al., 2021). We chose the batch size of 64 with maximum token length of 64, and use AdamW Optimiser (Bianchi et al., 2021), with cosine annealing learning rate. The best checkpoint was selected by minimum evaluation loss. All shared parameters are in the table below.

Hyperparameter	LoRA $r = 32$	DoRA $r = 32$	DoRA $r = 16$	AdaLoRA
Rank r	32	32	16	init=32, target=16
α	64	64	32	32
t_{init}	—	—	—	$0.10 \cdot T$
t_{final}	—	—	—	$0.50 \cdot T$
ΔT	—	—	—	10
β_1, β_2	—	—	—	0.85, 0.85
Orth. reg. weight	—	—	—	0.5
Trainable params	5.30M	5.44M	2.76M	$\sim 2.65\text{M}$

Table 2: Method-specific hyperparameters.

5 Results

5.1 Retrieval Performance and Baseline Comparison

Methods	Tuning Strategy	Trainable Params	MRR@1	MRR@5	MRR@10
NLLB-CLIP Base	Zero-Shot	0.00M	0.0218	0.0413	0.0462
NLLB-CLIP Large	Zero-Shot	0.00M	0.0248	0.0418	0.0480
AdaLoRA	target $r = 16$	2.65M	0.1337	0.2192	0.2374
DoRA	$r = 16$	2.76M	0.1912	0.2913	0.3093
DoRA	$r = 32$	5.44M	0.2115	0.3136	0.3332
LoRA	$r = 32$	5.30M	0.2214	0.3247	0.3441

Table 3: Text-to-Image retrieval performance across zero-shot baselines, and LoRA variants on the Burmese evaluation dataset.

Table 3 reports Mean Reciprocal Rank (MRR) at $k \in \{1, 5, 10\}$ on the held-out evaluation set of approximately 2,000 image-caption pairs on different methodologies. MRR is computed for text-to-image retrieval task meaning each Burmese caption is used to query and rank the 2,000-image evaluation dataset. LoRA at rank 32 achieves the strongest retrieval quality across all values of k , with MRR@10 exceeding 0.34, indicating that the correct image is retrieved within the top-10 candidates for more than one-third of Burmese text queries. DoRA at the same rank ($r=32$) performs similarly, which indicates that DoRA does not provide a meaningful advantage when adaptor capacity is sufficient (Liu et al., 2024). This is consistent with prior observations in literature, where DoRA’s gains are more observable at low rank (Han et al., 2024; Liu et al., 2024). DoRA with rank 16 retains approximately 86% of the LoRA ($r=32$) performance with a nearly 50% reduction in trainable parameters. This indicates that DoRA with lower rank is justifiable where storage, cost, and memory bandwidth constraints exist.

The worst-performing was AdaLoRA, achieving only 0.2374 MRR@10 despite using a similar number of parameters as DoRA ($r=16$). A deep dive reveals that query and key layers are heavily pruned in early, and middle layers (5,7, and 8) received near-zero rank allocations. This aggressive pruning may be detrimental in cross-modal settings where precise query-key interactions are essential for the alignment of text with visual embedding. This behaviour has been noted in the literature (Zhang et al., 2023a; Su et al., 2025). AdaLoRA consistently prefers to allocate more budget to FFN modules and top layers (Zhang et al., 2023a). Since query and key projections play a critical role in aligning representations across modalities, the aggressive pruning of attention layers suffer from sharp accuracy degradation (Su et al., 2025).

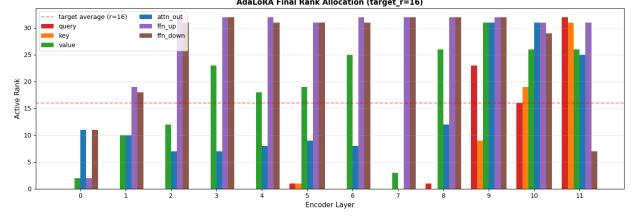


Figure 1: AdaLoRA Final Rank Allocation

5.2 Performance Comparison with NLLB-CLIP

The MRR for text-to-image retrieval was calculated for NLLB-CLIP Base and NLLB-CLIP Large under zero-shot condition (Visheratin, 2023b). Burmese-CLIP (LoRA, $r=32$) outperforms across all values of k , with MRR@10 improving from 0.0480 to 0.3441, approximately seven-fold increase. Whilst NLLB-CLIP’s text encoder possess Burmese language understanding, this result indicates that fine-tuning with a task-specific objective yields a better result, which is often the case in multilingual models that require explicit cross-modal supervision to achieve competitive retrieval performance (Wang et al., 2024). It is important to note that this comparison is against zero-shot NLLB-CLIP and therefore, the improvement could be attributed to task-specific fine-tuning itself rather than this specific methodology. A Burmese fine-tuned NLLB-CLIP would provide a stricter comparison.

5.3 Cross-Modal Alignment: Similarity Matrix

The Figure 2 presents a 10×10 cross-modal similarity matrix of best-performing model (LoRA $r = 32$) on a random sample of 10 image-caption pairs. It reports the cosine similarity between the text and image pairs. The diagonal entries achieve cosine similarities ranging from 0.44 to 0.82, with a mean diagonal score of approximately 0.62. The result demonstrates that meaningful cross-modal alignment has been established. Row labels in the figure are replaced with English captions for readability.

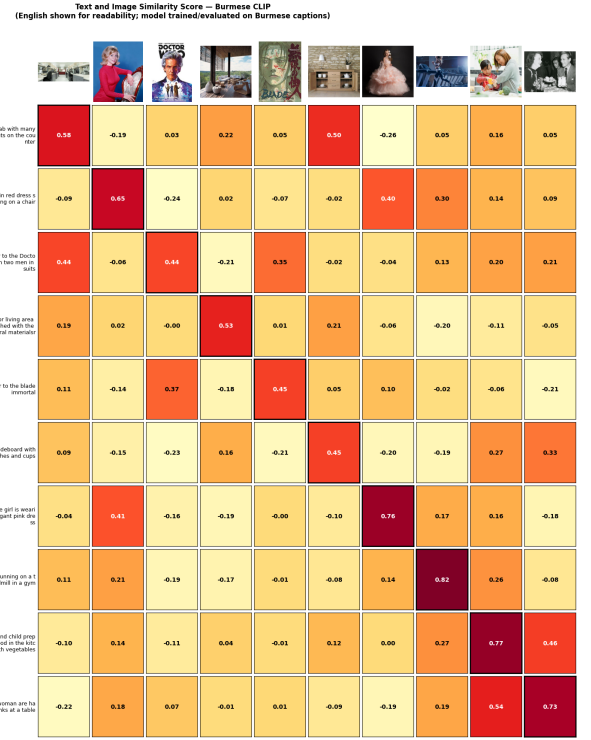


Figure 2: Similarity matrix (LoRA $r = 32$ model)

6 Limitations

First, the data used to train the model are machine-translated, which could cause semantic drift and translation artefacts. Second, the reported improvement over NLLB-CLIP is measured against a zero-shot baseline, and therefore, part of this gain may be attributable to task-specific fine-tuning itself rather than to the specific PEFT methodology evaluated here. A fine-tuned NLLB-CLIP baseline would offer a stricter point of comparison.

and is left for future work. Third, the results presented here represent a lower bound on what is achievable for Burmese vision–language modelling. A natural next step would be to curate a native Burmese image-text pair dataset to determine whether a better result can be achieved with full fine-tuning or PEFT-adapted models trained on non-machine-translated data. Fourth, a comparison of alternative text encoders should be studied to understand the impact of tokenisation choice on downstream performance, especially given the unique orthographic and segmentation challenges posed by the Burmese language. Fifth, the explanation of the poor performance by the AdaLoRA remains hypothetical since no ablation study was performed. Finally, the evaluation is restricted to a single contrastive retrieval task on a single language; the generalisation of these findings to other low-resource languages, or to non-contrastive objectives, remains an open question.

7 Conclusion

In this paper, we present the Burmese-CLIP trained on 20,000 machine-translated image-text pairs using various PEFT methods. The best Burmese-CLIP model, LoRA ($r=32$) outperformed Zero-Shot NLLB-CLIP Base and NLLB-CLIP Large on Mean Reciprocal Rank for text-to-image retrieval at $k \in \{1, 5, 10\}$ using the same held-out evaluation set of 2,000 image-text pairs. We also explored how other LoRA variants methodologies perform in the low-resource language setting. Across four LoRA variants evaluated under identical conditions, LoRA at rank 32 produced the strongest retrieval quality, suggesting that the LoRA formulation is often sufficient and does not require more complex adaptive variants to be effective in this setting. But the comparison also showed that alternative approaches are worth exploring. DoRA at rank 16 retained 86% of LoRA’s retrieval performance with roughly half the trainable parameters, offering a favourable efficiency trade-off for resource-constrained deployments, while AdaLoRA’s under-performance highlighted that adaptive methods can fail due to misallocation of ranks.

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